

The Role of Time, Space, and the Speed of Light

a companion piece to Minimal Necessary and Sufficient Conditions for a Physical Universe,

Third in the series of ΔQ : The Irreversible Universe

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Prologue

If everything happened everywhere, nothing could happen anywhere.

Introduction

It has long been the customary disposition of physical theory to regard space and time as primitive conditions within which physical processes unfold. Matter occupies position within space; events proceed in time; radiation propagates between them at a finite and invariant velocity denoted by c . Within this ordinary interpretation, the speed of light appears as a dynamical limitation imposed upon otherwise existing structure.

In the present work, this usual order of dependence is reversed.

Rather than treating finite propagation as a consequence of spacetime structure, let us consider whether spacetime itself becomes applicable only under conditions of finite propagation. The quantity c may then be understood not merely as a property of light, but as the invariant boundary preserving distinguishable relation among irreversible physical events.

In the preceding work, *Minimal Necessary and Sufficient Conditions for a Physical Universe*, it was argued that physical description first becomes applicable when a complementary distinction is irreversibly resolved. Such resolution incurs thermodynamic cost and establishes durable ordering. Yet durable ordering alone is insufficient. Were all irreversible distinctions instantaneously accessible to

one another, no meaningful separation between events could persist. Distinction would degenerate into global equivalence; locality would lose all purchase.

A universe admitting unrestricted instantaneous propagation is therefore not a universe of collapsed distances. It is a universe in which the relational structure underlying distance never arises. Under conditions of universal accessibility, irreversible distinctions remain ordered, yet no persistent separation can be maintained among them. Finite propagation prevents this collapse.

The quantity c therefore assumes a role more fundamental than that of a velocity. It establishes the invariant constraint governing how irreversible distinctions may be relationally shared without locality dissolving into global equivalence. In this sense, finite propagation does not merely limit a physical universe; it permits a universe of distinguishable relations to exist.

Light reveals this bound most directly, for massless propagation saturates the invariant limit without remainder. The significance of c , however, does not originate in light itself. Relativity reveals an underlying causal structure through light; the present work argues that this structure is prior to its manifestation.

From this perspective, space and time cease to appear as primitive containers. Time arises through the ordering imposed by irreversible distinction; space emerges as the minimal relational structure preserved by finite propagation among such distinctions. Spacetime is therefore not presupposed at the foundation of physical description, but becomes applicable only where irreversible distinction and finite propagation jointly obtain.

The purpose of the present work is to examine the consequences of this requirement and to clarify the structural relation between irreversibility, locality, finite propagation, and the applicability of spacetime description itself.

Irreversible Distinction and Durable Ordering

A complementary distinction provides the minimal condition under which meaningful definition may arise. Yet distinction alone remains abstract unless irreversibly resolved. Without irreversibility, ordering possesses no persistence; relations remain interchangeable and physically inapplicable.

Irreversible distinction stabilizes ordering by incurring durable thermodynamic commitment. In the present framework, this transition is bounded below by the thermodynamic conditions outlined by Landauer's principle and developed in the preceding work. The consequence of irreversible distinction is therefore not merely differentiation, but persistence.

Time arises from this persistence.

Temporal ordering is not introduced here as an independently existing background through which events proceed. Rather, time becomes applicable where irreversible distinctions admit durable relational sequence. Before such ordering exists, the distinction between prior and subsequent relation possesses no meaningful applicability.

Irreversibility therefore permits time by stabilizing distinguishable order.

Yet durable ordering alone remains insufficient to produce locality.

Finite Propagation and Relational Separation

If irreversible distinctions could be propagated instantaneously throughout the ordered structure, all relations would become universally accessible without delay. Under such conditions, no stable relational separation could persist from which locality or measurement might arise.

An instantaneous universe becomes relationally degenerate.

This degeneration does not signify the destruction of relation, but the collapse of distinguishable separation. Where all irreversible commitments are mutually adjacent, the distinction between “here” and “there” loses applicability. Spatial extension becomes undefined because no finite interval remains through which independent distinctions may remain structurally separated.

Finite propagation prevents universal synchronization.

The invariant quantity c therefore functions not merely as a velocity associated with radiation, but as the structural constraint preserving distinguishable locality among irreversible relations. It governs the conditions under which durable commitments may be consistently shared without collapsing into global equivalence.

In this sense, locality is not an auxiliary assumption imposed upon physical law. It is the structural consequence of constrained relational accessibility under finite propagation.

Without finite propagation, no stable relational separation exists from which measurement may arise.

The Applicability of Space

Durable ordering alone does not produce space. Space becomes applicable only where irreversible distinctions remain separated under finite propagation.

Space records the separation enforced by finite propagation.

Distance is therefore not fundamental in itself. It is descriptive. It expresses the constrained relational accessibility existing among irreversible events under the invariant propagation bound c .

Time provides durable order; finite propagation enforces locality; space records the resulting structure of separation.

The familiar geometry of spacetime emerges from these jointly necessary conditions. Spacetime is not the stage upon which irreversible processes unfold. It is the minimal relational structure becoming applicable once durable distinction and constrained propagation jointly obtain.

Under this interpretation, c does not generate spacetime dynamically, nor is it derived from electromagnetic theory alone. Rather, it appears as the invariant applicability condition required for distinguishable locality to persist without contradiction.

Light reveals this boundary because massless propagation saturates the invariant limit directly. Other physical processes fall below this limit due to additional relational constraints, not because the bound itself originates in their dynamics.

Invariance and Relativity

The invariance of c follows from the same structural necessity.

If the propagation bound governing irreversible distinction were contingent, frame-dependent, or locally variable, global synchronization could be reconstructed through comparative transformation. Distinguishable locality would again fail, as relational ordering could no longer remain consistently separated under transformation between observers.

The propagation bound must therefore remain universal and invariant if locality is to remain structurally preserved.

Relativity describes the kinematics of a universe already subject to this constraint. The present framework does not modify the operational predictions of relativity, nor reinterpret c as an emergent dynamical variable. Rather, it seeks to clarify why a constant possessing the role of (c) must exist at all.

Finite propagation preserves distinguishable locality. Relativity describes the resulting relational structure.

Falsifiability and Scope

The present framework concerns the conditions under which temporal and spatial description become meaningfully applicable, rather than the detailed dynamics governing systems within spacetime.

The framework would be falsified by any demonstration of durable physical distinction persisting without irreversible thermodynamic commitment, or by the existence of stable measurable relation absent finite propagation constraints.

Likewise, if irreversible distinctions could be globally synchronized without collapse of locality, the necessity of an invariant propagation bound would be undermined.

Furthermore, because the invariant speed limit c is introduced here as a necessary condition for preserving distinguishable locality among durable distinctions, any verified transfer of usable information that bypasses this propagation bound would render the emergent spacetime model untenable.

The framework does not replace relativity, thermodynamics, or quantum theory. It proposes the structural conditions under which their applicability becomes jointly coherent.

Conclusion

The present work argues that finite propagation is not merely a limitation imposed upon an already existing spacetime structure. It is the invariant condition permitting distinguishable locality to persist among irreversible relations.

Irreversibility permits durable temporal ordering. Finite propagation permits stable relational separation. Together, they permit the applicability of spacetime itself.

The quantity c therefore assumes a role deeper than that of a velocity associated with electromagnetic propagation alone. It defines the invariant boundary under which irreversible distinctions may remain both durably ordered and locally separable without collapsing into universal relational equivalence.

Space is not fundamental structure. It is the minimal relational framework licensed by irreversible distinction under finite propagation.

Under this interpretation, the speed of light appears not as a dynamical accident of nature, but as the operational revelation of a deeper structural necessity: the preservation of distinguishable relation within a universe of irreversible commitments.

Appendix A: Toy Model of Relational Accessibility

Consider only the logical accessibility among a set of irreversible distinctions:

$$\mathbf{D_1, D_2, D_3, D_4}$$

Each distinction is durable once resolved. The question is whether these distinctions can remain locally distinguishable.

Case 1: Instantaneous Accessibility

If every distinction becomes accessible to every other distinction with no propagation interval, then:

$$\{\mathbf{D}_1, \mathbf{D}_2, \mathbf{D}_3, \mathbf{D}_4\} \rightarrow \mathbf{D}_{1234}$$

without relational interval.

No distinction remains locally isolated. No interval separates one relation from another. The system no longer preserves ordered locality, but a single globally adjacent relational state:

$$\mathbf{D}_{1234}$$

In such a model, “here” and “there” possess no operational meaning because every relation is mutually accessible without delay.

Case 2: Finite Accessibility

Now impose a finite propagation rule:

$$\mathbf{D}_1 \rightarrow \mathbf{D}_2 \rightarrow \mathbf{D}_3 \rightarrow \mathbf{D}_4$$

Each distinction becomes accessible only through constrained relation. Now there exists an interval during which:

has affected

$\mathbf{D}_1$

$\mathbf{D}_2$

but not yet

$\mathbf{D}_3$

$\mathbf{D}_4$

This permits locality.

The structure now supports:

$$\mathbf{here} \neq \mathbf{there}$$

Finite propagation is not added to an already spatial system. Rather, finite propagation is the condition under which spatial separation becomes meaningfully definable.

This appendix is illustrative rather than dynamical and is intended only to clarify the structural distinction between constrained and unrestricted relational accessibility.